

Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects*

Liyang Sun[†] and Sarah Abraham[‡]

September 22, 2020

Abstract

To estimate the dynamic effects of an absorbing treatment, researchers often use two-way fixed effects regressions that include leads and lags of the treatment. We show that in settings with variation in treatment timing across units, the coefficient on a given lead or lag can be contaminated by effects from other periods, and apparent pretrends can arise solely from treatment effects heterogeneity. We propose an alternative estimator that is free of contamination, and illustrate the relative shortcomings of two-way fixed effects regressions with leads and lags through an empirical application.

Keywords: difference-in-differences, two-way fixed effects, pretrend test

*We are grateful to Isaiah Andrews, Amy Finkelstein, Anna Mikusheva, and Heidi Williams for their guidance and support. We thank Alberto Abadie, Jonathan Cohen, Nathan Hendren, Peter Hull, Guido Imbens, Yunan Ji, Sylvia Klosin, Kevin Kainan Li, Paichen Li, Therese A. McCarty, Whitney Newey, James Poterba, Pedro H. C. Sant'Anna, Gergely Ujhe-lyi and Helen Willis for helpful discussions. This research was supported by the National Institute on Aging, Grant Number T32-AG000186. A preliminary draft of this paper was circulated on April 16, 2018. Replication code is available at <http://economics.mit.edu/grad/lsun20/>. The companion Stata package `eventstudyweights` is available from the SSC repository.

[†]Department of Economics, MIT, 77 Massachusetts Avenue, Cambridge, MA 02139. Corresponding author; lsun20@mit.edu.

[‡]Cornerstone Research, 699 Boylston St, Boston, MA 02116. The views expressed herein are solely those of the author, who is responsible for the content, and do not necessarily represent the views of Cornerstone Research.

following differences. First, in the setting where there is no never-treated group, our method uses the last cohort to be treated as a control group, whereas Callaway and Sant’Anna (2020a) use the set of not-yet-treated cohorts. Our method and theirs thus rely on different, but non-nested parallel trends assumptions. Second, our estimation method can be cast as a regression specification and thus may be more familiar to applied researchers. However, a third difference is that the procedure of Callaway and Sant’Anna (2020a) allows for conditioning on time-varying covariates. de Chaisemartin and D’Haultfœuille (2020) and Goodman-Bacon (2018) respectively propose alternative estimators and diagnostic tools for estimation of causal effects in staggered settings, but do not consider the estimation of the dynamic path of treatment effects as we do.

2 Event studies design

In this section we first formalize the “event studies design”. As discussed in Section 2.3, based on how this term is deployed in the empirical literature, an *event study* design is a *staggered adoption* design where units are treated at different times, and there may or may not be never treated units. It also nests a *difference-in-differences* design, where units are either first treated at time t_0 or never treated.

Specifically, we consider a setting with a random sample of N units observed over $T + 1$ time periods, where T is fixed. For each $i \in \{0, \dots, N\}$ and $t \in \{0, \dots, T\}$, we observe the outcome $Y_{i,t}$ and treatment status $D_{i,t} \in \{0, 1\}$: $D_{i,t} = 1$ if i is treated in period t and $D_{i,t} = 0$ if i is not treated in period t . Throughout we assume that the observations $\{Y_{i,t}, D_{i,t}\}_{t=0}^T$ are independent and identically distributed (i.i.d.).

In the general case of event studies we focus on an *absorbing* treatment such that the treatment status over time is a non-decreasing sequence of zeros and then ones, i.e. $D_{i,s} \leq D_{i,t}$ for $s < t$. We can thus uniquely characterize a treatment path by the time period of the initial treatment, denoted with $E_i = \min \{t : D_{i,t} = 1\}$. If unit i is never treated i.e. $D_{i,t} = 0$ for all t , we set $E_i = \infty$. Based on when they first receive the treatment, we can also uniquely categorize units into disjoint cohorts e for $e \in \{0, \dots, T, \infty\}$, where units in cohort e are first treated at the same time $\{i : E_i = e\}$.

We define $Y_{i,t}^e$ to be the potential outcome in period t when unit i is first treated in time period e . We define $Y_{i,t}^\infty$ to be the potential outcome if unit i never receives the treatment, which we call the “baseline outcome”. Since the timing of the initial treatment uniquely characterizes one’s treatment path, we can

Assumption 3 requires that each cohort experiences the same path of treatment effects. Treatment effects need to be the same across cohorts in every relative period for homogeneity to hold, whereas for heterogeneity to occur, treatment effects only need to differ across cohorts in one relative period. The assumption of treatment effect homogeneity is therefore strong, and in Section 3.4.1, we describe how it can be violated in applied settings.

Our notion of treatment effect homogeneity does not preclude dynamic treatment effects; it only imposes that cohorts share the same path of treatment effects. The related literature sometimes formulates restrictions on the dynamics of treatment effects as another notion of treatment effect homogeneity. Athey and Imbens (2018) propose an assumption that “restricts the heterogeneity of the treatment effects over time,” which implies $CATT_{e,\ell}$ can vary over e but not over ℓ . Borusyak and Jaravel (2017) refer to one type of treatment effects heterogeneity as “only across the time horizon,” which implies $CATT_{e,\ell}$ can vary over ℓ but not over e . Callaway and Sant’Anna (2020a) allow for “arbitrary treatment effect heterogeneity” when $CATT_{e,\ell}$ varies across cohorts and over time. Similarly, de Chaisemartin and D’Haultfœuille (2020) describe treatment effects that may be “heterogeneous across groups and over time periods.” Goodman-Bacon (2018) allow heterogeneous effects to either “vary across units but not over time” or “vary over time but not across units.” The literature has not converged on a single notion of treatment effects heterogeneity with time-varying treatment. Since researchers are interested in dynamic treatment effects when using a “dynamic” specification, we do not restrict the path of treatment effects but rather use “heterogeneity” to describe variation across cohorts only.

2.3 Relevance in the applied literature

To gauge the empirical relevance of our results, we survey the estimation methods used by the twelve papers collected by Roth (2019) from three leading economics journals that contain the phrase “event study” in their main text.¹ From this sample of applied papers, we learn what specifications empirical researchers are actually using when estimating two-way fixed effects regressions. Four papers in this sample consider the simple setting where units either receive their first treatment at the same time or never receive the treatment. The other eight papers in this sample consider the more complex setting where treated units receive their

¹We follow the selection criteria in Roth (2019): the original sample consists of 70 total papers, but is further constrained to these twelve papers with publicly available data and code. The data and code are used to determine exactly the specification estimated in these papers.

treatment at various times, and there may or may not be never treated units. This observation suggests an *event study* in the applied literature nests two popular research designs: *difference-in-differences* design, but also the design where units receive their first treatments at various times, which is our focus.²

We summarize the main specifications in this sample of twelve applied papers in Table 1.³ The columns of Table 1 collect key properties of these specifications. In Section 3.1, we introduce a general class of specifications that encompasses all of these specification. The estimates for relative period coefficients μ_ℓ from all these papers therefore fall under our analysis in the next section.

These papers demonstrated that event studies are used to address a broad range of research questions. As an example of this literature, Bailey and Goodman-Bacon (2015) use the rollout of the first Community Health Centers (CHCs) to study the longer-term health effects of increasing access to primary care. As another example, Tewari (2014) uses variation in the timing of deregulation across states to estimate the impact of financial development on homeownership.

3 Estimators from linear two-way fixed effects regression

We consider a two-way fixed effects (FE) regression of the following form, estimated on a panel of $i = 1, \dots, N$ units for $t = 0, 1, \dots, T$ calendar time periods:

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{g \in \mathcal{G}} \mu_g \mathbf{1}\{t - E_i \in g\} + v_{i,t} \quad (4)$$

Here $Y_{i,t}$ is the outcome of interest for unit i at time t , E_i is the time for unit i to initially receive a binary absorbing treatment, and α_i and λ_t are the unit and time fixed effects. The set $\mathcal{G}$ collects disjoint sets g of *relative periods* $\ell \in [-T, T]$. We allow some relative periods to be excluded from the specification and denote the excluded set with $g^{excl} = \{\ell : \ell \notin \bigcup_{g \in \mathcal{G}} g\}$. We denote by μ_g the relative period coefficients from regression (4), i.e. the population regression coefficients. Their corresponding OLS estimators are denoted by $\hat{\mu}_g$ respectively.

We are interested in the properties of μ_g when there are variations in the initial treatment timing, and

²This setup is the same as the *staggered adoption* design proposed by Athey and Imbens (2018), but we keep the term *event study* because it is common in the applied literature.

³We focus on the first specification underlying the event study estimates in each paper, which we view as a reasonable proxy for the main specification in the paper.

relative periods $\ell' \notin g$, with the same weights stated in Proposition 1:

$$\mu_g = \sum_{\ell \in g} \sum_e \omega_{e,\ell}^g CATT_{e,\ell} + \sum_{g' \neq g, g' \in \mathcal{G}} \sum_{\ell' \in g'} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{\ell' \in g^{excl}} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'}. \quad (13)$$

Under Assumption 1, the terms in Proposition 1 reduce to a linear combination of the causally interpretable building blocks $CATT_{e,\ell}$ as follows:

$$E[Y_{i,e+\ell} - Y_{i,0}^\infty | E_i] - E[Y_{i,e+\ell}^\infty - Y_{i,0}^\infty] = CATT_{e,\ell} + \underbrace{E[Y_{i,t}^\infty - Y_{i,0}^\infty | E_i] - E[Y_{i,t}^\infty - Y_{i,0}^\infty]}_{=0} \quad (14)$$

for $t = e + \ell$. However, two issues for interpretability remain. First, the coefficient μ_g can be written as an average of not only $CATT_{e,\ell}$ from own periods $\ell \in g$, but also $CATT_{e,\ell'}$ from other periods. Second, the weights are still non-linear functions of the distribution of cohorts, same as those in (12), and they are not restricted to lie in $[0, 1]$.

The properties of these weights as described following Proposition 1 imply that contamination from other periods wanes once we impose restrictions on treatment effects. In the next two subsections we illustrate how that can happen.

3.4 Interpreting the coefficients under parallel trends and no anticipation assumptions

Proposition 3. *If Assumption 1 (parallel trends) holds and Assumption 2 (no anticipatory behavior in all periods before the initial treatment) holds, the population regression coefficient μ_g is a linear combination of post-treatment $CATT_{e,\ell'}$ for all $\ell' \geq 0$, with the same weights stated in Proposition 1:*

$$\mu_g = \sum_{\ell' \in g, \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{g' \neq g, g' \in \mathcal{G}} \sum_{\ell' \in g', \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'} + \sum_{\ell' \in g^{excl}, \ell' > 0} \sum_e \omega_{e,\ell'}^g CATT_{e,\ell'}. \quad (15)$$

Once we restrict pre-treatment $CATT_{e,\ell \leq 0}$ to be zero under the no anticipatory behavior assumption, the expression for μ_g simplifies as terms involving $CATT_{e,\ell \leq 0}$ drop out. However, the second term in the expression for μ_g remains unless we further impose treatment effect homogeneity for its summands to cancel out each other. Thus, μ_g may be non-zero for pre-treatment periods even if parallel trends holds.

This result immediately implies a shortcoming of using pre-treatment coefficients (i.e. μ_g where g contains only leads to the treatment $\ell < 0$) to test for pretrends. Under the no anticipatory behavior assumption,

cohort-specific treatment effects prior to treatment are all zero: $CATT_{e,\ell} = 0$ for all $\ell < 0$. Therefore, any linear combination of these $CATT_{e,\ell}$ is also zero. However, μ_g is a function of post-treatment $CATT_{e,\ell' \geq 0}$ as well, even when g only contains elements with $\ell < 0$. We revisit this implication in greater depth in Section 3.7. Callaway and Sant’Anna (2020a) provides alternative tests for pretrends that do not suffer from this drawback.

3.4.1 Sources of treatment effect heterogeneity

Since treatment effects heterogeneity violates Assumption 3 and can alter how we interpret μ_g , it is important to think through when different cohorts likely experience different paths of treatment effect. Such heterogeneity could arise for many reasons. For example, cohorts may differ in their covariates, which affect how they respond to treatment. We will explore a concrete example in our application: if treatment effects differ with age, and there is variation in age across units first treated at different times, we will have heterogeneous effects (see Section 5 for details). After controlling for covariates, cohorts may still vary in their responses to the treatment if units select their initial treatment timing based on treatment effects. This source of heterogeneity is still compatible with our parallel trends assumption, which only rules out selection in the initial treatment timing based on the evolution of the baseline outcome. In addition to these two sources of heterogeneity, treatment effects may vary across cohorts due to calendar time-varying effects (e.g. macroeconomic conditions could govern the effects on labor market outcomes across cohorts).

3.5 Interpreting the coefficients under parallel trends and treatment effect homogeneity

Proposition 4. *If Assumption 1 (parallel trends) holds and Assumption 3 (treatment effect homogeneity) holds, then $CATT_{e,\ell} = ATT_\ell$ is constant across e for a given ℓ , and the population regression coefficient μ_g is equal to a linear combination of $ATT_{\ell \in g}$, as well as $ATT_{\ell' \notin g}$ from other relative periods:*

$$\mu_g = \sum_{\ell \in g} \omega_\ell^g ATT_\ell + \sum_{g' \neq g} \sum_{\ell' \in g'} \omega_{\ell'}^g ATT_{\ell'} + \sum_{\ell' \in g^{excl}} \omega_{\ell'}^g ATT_{\ell'} \quad (16)$$